

AI-Based Fish Species Classification Using Phenotypic Features

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Abstract—The accurate identification of fish species is an important endeavour in the study of marine biodiversity and underwater ecosystems. This paper presents a hybrid classification framework that integrates handcrafted phenotypic and deep learning-based features. The resultant classification framework utilizes images obtained from the Fish4Knowledge dataset and uses masks to ensure the extraction of features only from the fish areas within the images. The masks were used to segment the photographs into fish and non-fish sections, enabling the extraction of morphologic and spatial characteristics (shape, texture, colour) along with six pretrained CNNs (VGG16, ResNet50, MobileNetV3, Xception, EfficientNetB0, InceptionV3) from deep neural networks. Subsequently, all the extracted features were subjected to dimensionality reduction using PCA and were evaluated by the following classifiers: SVM, Random Forest and XGBoost. The highest accuracy was achieved with the combination of handcrafted features and the Xception embeddings, which provided an overall classification accuracy of 97.42% when combined with an SVM classifier. When compared to other studies, the hybrid approach described in this paper has shown to be competitive in providing a practical, effective methodology for classifying underwater fish species.

Keywords—*Fish phenotype classification, Fish4Knowledge dataset, U-Net segmentation, Handcrafted features, Deep CNN features, Feature fusion, Principal Component Analysis (PCA), Support Vector Machine (SVM), Random Forest, XGBoost.*

I. INTRODUCTION

Automated fish species identification is crucial for assessing marine biodiversity and monitoring fisheries and ecosystem health [1], [2]. Traditional manual monitoring methods are often inefficient, labor-intensive, and prone to human error, particularly when applied to large-scale datasets [3]. Fortunately, rapid improvements in underwater imaging and com-

puter vision have made automated approaches increasingly feasible. Past research has primarily focused on two distinct approaches: handcrafted feature descriptors (such as shape, color, and texture) and deep learning representations [4]. Handcrafted features provide interpretability but often fail to capture the challenges of underwater environments, such as variable illumination, occlusion, and complex backgrounds [3], [5]. Conversely, deep learning models—specifically Convolutional Neural Networks (CNNs)—have demonstrated remarkable performance but typically require enormous, annotated datasets to generalize well [1], [6]. In this study, we propose a hybrid framework that synergizes handcrafted phenotypic traits with deep feature extraction. Utilizing 27,358 images across 23 species from the Fish4Knowledge dataset [6], our approach is distinguished by a rigorous U-Net-based segmentation stage prior to feature extraction [7], [8]. We produced accurate masks for each fish to isolate regions of interest, enabling the accurate calculation of shape, color, and texture features. Simultaneously, we extracted deep embeddings from six pretrained CNN architectures and reduced their dimensionality using Principal Component Analysis (PCA). By combining these representations and testing them on Support Vector Machine (SVM), Random Forest, and XGBoost classifiers, we show that a hybrid approach produces a more robust and accurate system for underwater species identification.

II. RELATED WORK

In the past few decades, many researchers have developed methods of fish species identification to facilitate various applications such as aquaculture, underwater surveillance, and ecosystem assessment. Earlier studies relied heavily on handcrafted features like shape descriptors, color histograms, Local

Binary Patterns, and GLCM. While these were useful to capture body structure and texture patterns, their performance fell significantly in underwater conditions due to challenges such as poor lighting, turbidity in water, and complex backgrounds. [3], [4]

With advancements in deep learning, CNN-based models have recently become very popular since they learn discriminative patterns directly from images without the need for manual feature design. Strong performance of some powerful deep networks, including VGG, ResNet, Xception, and EfficientNet, has been reported in fish classification-especially in visually degraded underwater conditions. However, the major problem with CNNs is that they usually require a large labeled dataset with high computational requirements, making it difficult to deploy them in real-time aquaculture systems. [1], [2], [5]

Several recent works have sought a balance between accuracy and interpretability in hybrid models combining handcrafted features with deep learning-based ones. Handcrafted features provide transparency and capture biologically meaningful phenotypic traits, while deep features capture high-level structures, spatial relationships, and fine-grained variations across species. Studies in both marine ecology and underwater imaging demonstrate that integrating these feature types can improve the stability and robustness of predictions, even in noisy underwater conditions. [6], [7]

Research focused on segmentation has also emphasized the need to isolate fish regions from complex underwater backgrounds by employing architectures like the U-Net, LCFCN, and ResNet-FCN, improving the dependability of downstream feature extraction greatly. [8], [9], [10]

Although these contributions have, in this way, enhanced the performance in fish classification, most of the existing methods either rely exclusively on CNNs or do not perform an explicit analysis of the contribution of various feature sets toward final results. The goal of this work is to fill this gap by a systematic evaluation of handcrafted, deep, and hybrid features across multiple classifiers, showing that a combination of traditional phenotypic traits with deep learning features provides a more reliable approach toward identifying underwater species.

III. METHODOLOGY

After identifying the fish, two complementary methods were used to extract features: handcrafted phenotypic descriptors and deep CNN embeddings. Handcrafted features were calculated from the fish mask itself and are represented in shape-based measurements (e.g., aspect ratio, area, perimeter, circularity and Hu moments). Handcrafted features reflect common characteristics used by marine biologists to identify fish species. To capture textural characteristics, texture information was calculated using LBP and GLCM statistics, and colour characteristics were captured using RGB and HSV histograms that reflect natural pigment patterns.

Hybrid features were derived from six pretrained CNN models (VGG16, ResNet50, MobileNetV3L, Xception, EfficientNetB0, and InceptionV3). The features generated from these networks were able to capture subtle morphological

characteristics and texture differences that may not be present using handcrafted features alone.

All feature vectors were subsequently reduced in dimensionality through PCA. The generated reduced feature sets were combined and termed a hybrid descriptor, as it exploits the advantages of both handcrafted features and deep CNN-based feature extraction.

A. Description of the Dataset

The dataset employed in this research is the Fish4Knowledge dataset, comprising 27,358 fish images and 23 species, together with their binary masks of corresponding 27,358 images, amounting to a total of 54,716 images. The dataset was captured under real-world conditions and thus has challenges such as turbidity, low visibility, occlusion, and cluttered backgrounds. The dataset has augmentations for generalization, such as random rotation, flip, zoom, and contrast change. This dataset thus provides a real-world benchmark for testing classification performance in aquaculture

B. Data Preprocessing and Segmentation

The underwater images tend to have cluttered backgrounds, which can negatively impact the classification accuracy as shown in Fig. 1. To overcome this, our system includes an automated segmentation process. Although the Fish4Knowledge dataset offers ground-truth references, we developed a U-Net model to obtain the binary masks required in our feature extraction process. The encoder-decoder model allows the network to focus on both high-resolution spatial information and deep semantic information to achieve a precise localization of the fish body. The resulting masks are then used as stencils to extract the phenotypic characteristics, such as shape and texture, from the background.

1) *U-Net Based Segmentation*: The first part of the process involves a U-Net model that carries out the pixel-level segmentation of fish images, resulting in the creation of accurate binary masks which distinguish the fish bodies from their watery background shown in Fig. 2. The encoder-decoder architecture of U-Net, along with the skip connections, gets the semantic features from the deep layers and combines them with the high-resolution spatial details so as to get an accurate localization. The encoder reduces the input area step-by-step so as to capture the contextual information, while the decoder increases the input area step-by-step so as to rebuild the segmentation map with very precise boundary cutting. This segmentation process is very important in underwater environments, as the turbidity of the water, background clutter, and changing lighting conditions very often cause the fish to be barely seen.

C. Feature Extraction

Feature Extraction Methods: Following the mask generation phase, features are extracted using classical approaches and deep approaches. Classical strategies yield phenotypic descriptors, such as shape (length, width, aspect ratio) and

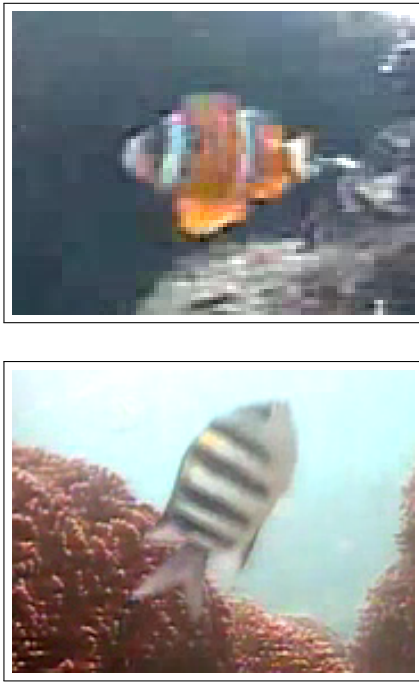


Fig. 1. Sample fish images from the dataset used.

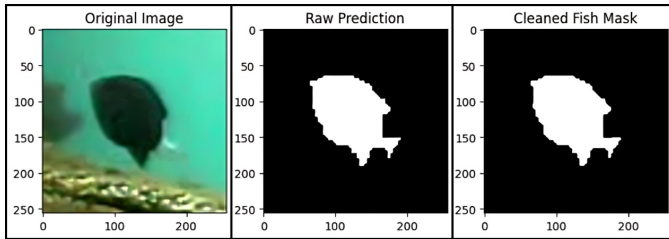


Fig. 2. Sample input image and its generated mask

texture (GLCM, LBP), plus color histograms that are more interpretable biological markers that experts routinely use to identify species. At the same time, deep embeddings from pretrained networks yield higher-level signatures of size, deformations, and subtle pattern differences that handcrafted descriptors may miss. Principal Component Analysis (PCA) is then used to reduce redundancy and aid in selecting the most discriminative information across the classical and deep feature domains. This dimensionality reduction provides a more computationally-efficient feature set, reduces robustness against variation, and tolerates minor environmental variability while capturing biologically relevant cues, thus reducing the feature set to a compact and discriminative classification feature set.

Feature extraction was performed in four separate categories to capture morphological, textural, color, and high-level semantic information from the fish dataset:

- 1) **Shape Features (Mask only):** Morphological features, including body length, width, area, perimeter, aspect ratio, circularity, and Hu moments, were computed from the binary masks of the segmented fish.

- 2) **Texture and Color Features (Image + Mask as stencil):** Features derived from the segmented fish regions included Local Binary Patterns (LBP), Gray-Level Co-occurrence Matrix (GLCM) features, and histograms in both RGB and HSV color spaces.
- 3) **Deep Features (Image only):** The dataset was used to make slight adjustments to the six pre-trained convolutional neural network (CNN) architectures VGG16, ResNet50, MobileNetV3L, Xception, EfficientNetB0, and InceptionV3, so that they could be responsible for the extraction of high-level semantic and morphological representations.
- 4) **Combined Features:** To improve the distinguishing power for classification, a complete feature vector was created through the combination of manual (shape, texture, color) and deep features.

The structured feature sets were output as CSV files to be usable in subsequent analyses and classification tasks.

D. Classifier Models

To evaluate the extracted features, three machine learning classifiers were employed:

- 1) **Support Vector Machine (SVM):** Implemented with a radial basis function (RBF) kernel to handle non-linear decision boundaries in high-dimensional feature spaces.
- 2) **Random Forest (RF):** A combination of decision trees trained using bootstrap aggregation to enhance robustness against noisy data and diverse features.
- 3) **XGBoost:** A gradient boosting library optimized using regularization techniques, famous for its high predictability and scalability in structured feature spaces.

Each classifier was tested separately using four different feature sets: shape, texture and color, deep features, and combined features. Additionally, the CNN models were evaluated based on their learned deep embeddings.

The comparative analysis of the classifiers demonstrates a trade-off between accuracy, interpretability, and scalability. Ensemble classifiers perform well in terms of generalization, and SVM provides excellent classification results, especially in cases involving complex decision boundaries, thereby making it a reliable method for fish species classification.

E. Implementation Details

The proposed framework was implemented in Python using TensorFlow/Keras for training and extracting features via CNN, OpenCV and NumPy for handcrafted feature extraction, and scikit-learn and XGBoost for classification. The training and testing process was accelerated by an NVIDIA GPU, while handcrafted feature extraction and classification were performed on CPU to simulate lightweight deployment. The structured evaluation with four series of characteristics, three different classifiers, and six different CNN models offers a broad insight into the trade-offs between interpretability, complexity, and accuracy.

F. Training and Evaluation

The classifiers such as XGBoost, Random Forest, and SVM were trained on the hybrid feature sets that were obtained after the segmentation and feature extraction phase. The deep CNN embeddings such as ResNet50 and EfficientNet were first pre-trained on the ImageNet dataset and then fine-tuned using a tuning process that utilized the cross-entropy loss function and the Adam optimizer (with a learning rate of 0.0001). The batch size was fixed at 32, and the models were trained for a period of 20-30 epochs to enable a stable learning process and avoid overfitting. The performance of the models was extensively evaluated using a variety of metrics. Test accuracy was the primary indicator, and precision, recall, and F1-score gave insights into class-wise effectiveness. A confusion matrix was used to depict misclassifications among species. All these metrics combined gave a strong validation of the system under different underwater conditions.

IV. RESULTS AND DISCUSSION

The research examines the identification of various fish species using handcrafted features based on phenotype and classification based on deep learning embedding. We examined morphological, color, and texture descriptors using traditional classifiers, and we also examined deep features using six pretrained CNNs, both individually and fused with others. Using the open dataset Fish4Knowledge (23 species, 27,358 images with masks), we offered a comprehensive side-by-side comparison of all examined feature sets and classifiers and provide insight into the unique pros and cons of each to facilitate efficient and accurate fish identification.

A. Performance of Handcrafted Features

There was a huge difference in the discriminating ability of phenotypic features: Table I shows that among the different features shape descriptors were the least performing, and the SVM, RF, and XGBoost classifiers made 55%, 62%, and 63% accuracy respectively, which shows that shape feature alone is not sufficient as there are similar silhouettes and pose variations; texture features helped to increase accuracy, with SVM at 83%, RF at 80%, and XGBoost at 82%, which indicates that surface patterns are indeed sensitive to lighting and hence still provide useful cues; while color proved to be the most informative feature with SVM at 90%, RF at 92%, and XGBoost at 94%, thus it is indeed possible to determine by means of pigmentation patterns which species are being referred to. In conclusion, color is the most important among the manually selected features, texture has a moderate power to discriminate, and shape by itself cannot be depended upon to classify species at the level of uniqueness.

B. Deep Feature Evaluation

Table II shows that Deep embeddings extracted from six CNN models (VGG16, ResNet50, MobileNetV3L, Xception, EfficientNetB0, InceptionV3) demonstrated superior classification capabilities. Individually, EfficientNetB0 and Xception produced the highest accuracies (≈ 96 –97%), while lighter

TABLE I
MODEL ACCURACY COMPARISON ACROSS ALL FEATURE SETS

Feature Set	SVM	Random Forest	XGBoost
Color Features	0.9012	0.9253	0.9362
Shape Features	0.5499	0.6208	0.6323
Texture Features	0.8307	0.8043	0.8179

models like MobileNetV3 and VGG16 were computationally efficient but slightly less accurate. The results underscore the effectiveness of CNN embeddings in capturing complex intra-class variations and robust underwater features.

TABLE II
MODEL ACCURACY COMPARISON ACROSS ALL DEEP FEATURES

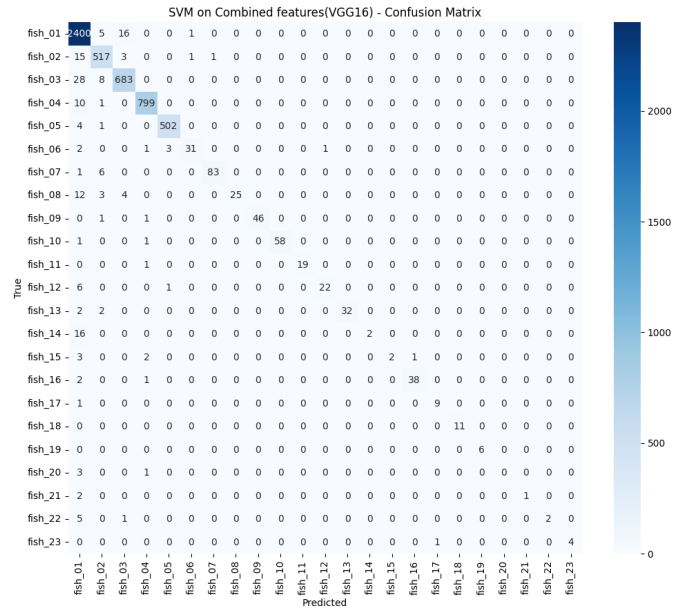
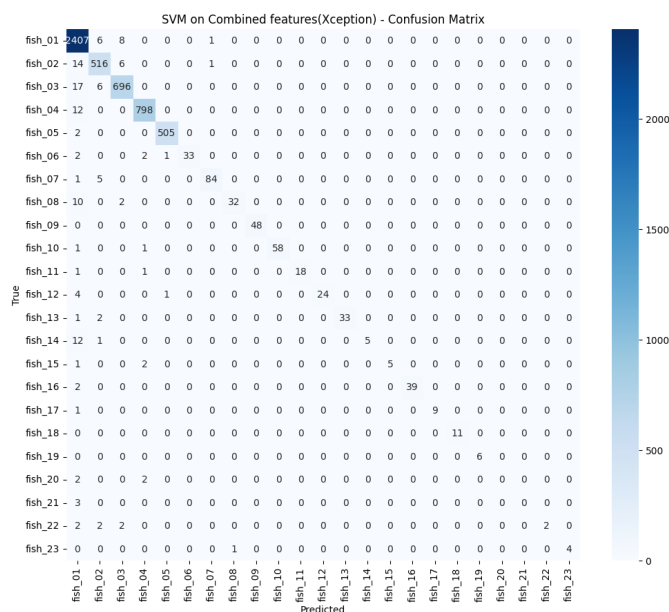
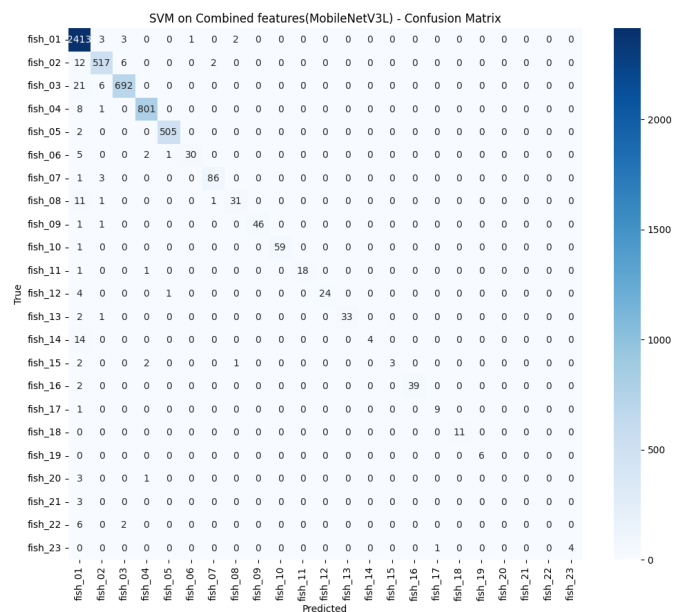
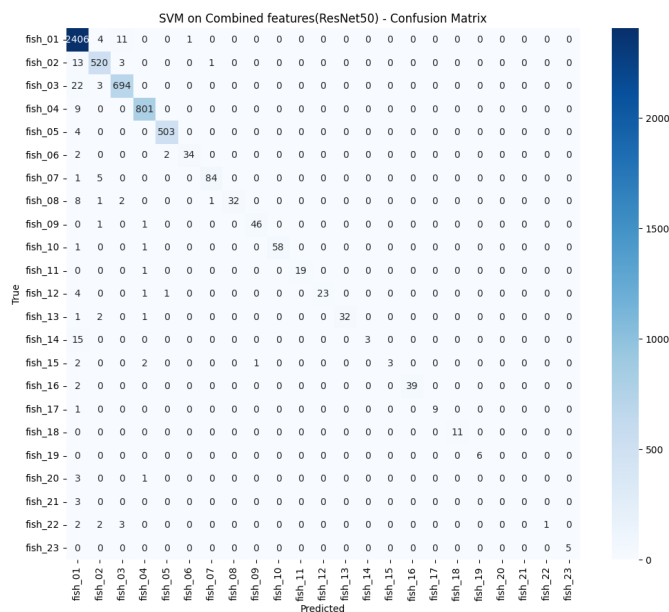
Models	SVM	Random Forest	XGBoost
EfficientNetB0	0.9715	0.9251	0.9589
InceptionV3	0.9688	0.9214	0.9582
MobileNetV3L	0.9739	0.9399	0.9624
ResNet50	0.9735	0.9311	0.9602
VGG16	0.9668	0.9220	0.9534
Xception	0.9742	0.9291	0.9600

C. Performance Analysis of Combined Feature Sets

The combined feature sets that incorporated hand-crafted descriptors of shape, color, and texture and deep CNN embeddings always produced high classifier accuracy for all of the models. Of the different configurations tested, the combined features and the Xception model had the highest accuracy of 0.9742 when evaluated using the support vector machine classifier. The combined features and the MobileNetV3L model had the second highest results with 0.9739 accuracy, followed close to the combined features and ResNet50 with an accuracy of 0.9735, and finally, the combined feature set with EfficientNetB0 was also high in accuracy with 0.9715 accuracy as shown in Table III. The results support the roles of handmade features and CNN representation in increased discriminative power, combining the features is more powerful instead of using one or the other. Additionally, the findings also demonstrate that some deep models, such as the Xception and MobileNetV3L, provide embeddings that compliment hand-crafted features best while also providing more accurate classifications. Overall, the results support the findings that this hybrid feature approach is robust for fish phenotype recognition tasks.

TABLE III
ACCURACY OF CLASSIFIERS ON COMBINED FEATURES (HANDCRAFTED + DEEP)

Combined Set	SVM (%)	RF (%)	XGBoost (%)
Handcrafted + VGG16	96.68	92.20	95.34
Handcrafted + ResNet50	97.35	93.11	96.02
Handcrafted + MobileNetV3L	97.39	93.99	96.24
Handcrafted + Xception	97.42	92.91	96.00
Handcrafted + EfficientNetB0	97.15	92.51	95.89
Handcrafted + InceptionV3	96.88	92.14	95.82



D. Discussion

The results suggest that hybrid phenotypic–deep learning pipelines strike an effective and practical balance between accuracy, efficiency, and interpretability, with handcrafted features containing all biologically meaningful cues, and deep embeddings capturing complex visual patterns. Class-wise analyses evidence that species with high visual similarity gain the most advantage from feature fusion. These results support hybrid methods for ecological monitoring, and indicate that the presented framework is suitable for real-world underwater

deployment to improve biodiversity assessments and smart aquaculture management.

E. Computational Complexity and Real-Time Feasibility

Practical implementation in aquaculture requires a fine-tuned balance between classification precision and processing latency. While the Xception-based hybrid configuration achieved the peak accuracy of 97.42%, the MobileNetV3 model reached a comparable 97.39% accuracy (see TABLE III) with a significantly smaller parameter footprint, positioning it as the more efficient candidate for memory-constrained

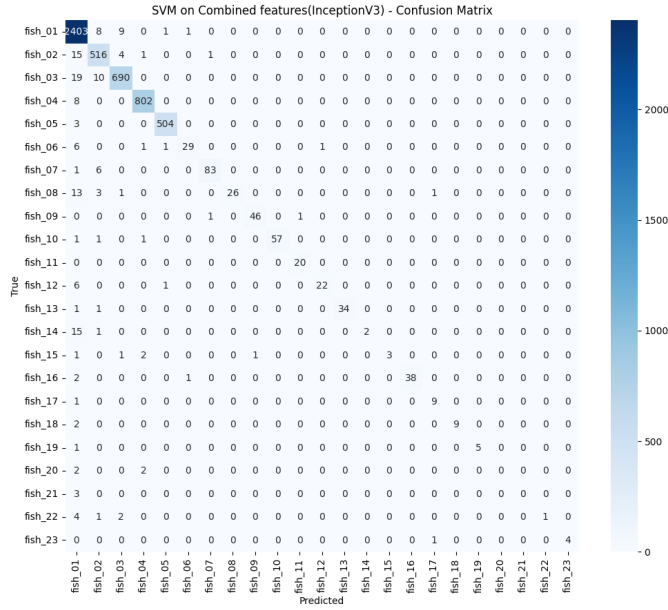


Fig. 7. Confusion Matrix for SVM on Combined Features- InceptionV3.

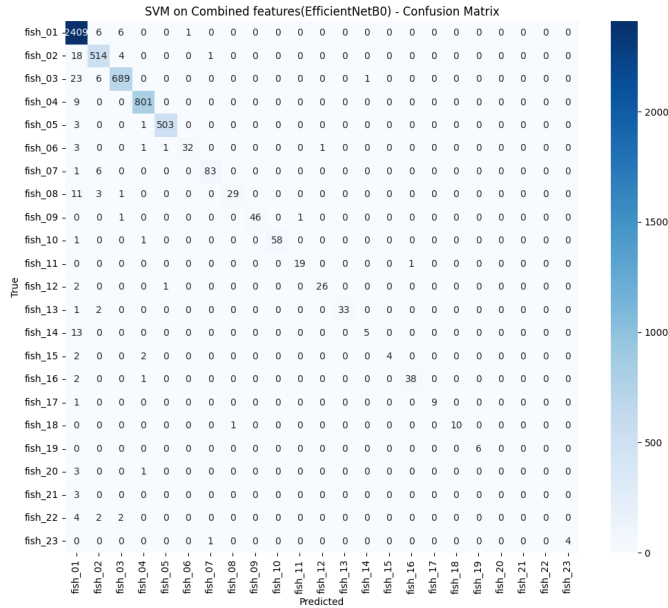


Fig. 8. Confusion Matrix for SVM on Combined Features- EfficientNetB0.

edge hardware. To simulate realistic deployment, handcrafted feature extraction was offloaded to the CPU while utilizing GPU acceleration for deep embeddings. This pipeline was further optimized using Principal Component Analysis (PCA), which reduced the high-dimensional hybrid vectors into a smaller set of discriminative features. This helped in reducing the computational complexity and time taken for the final classification task without affecting the integrity of the system.

V. CONCLUSION

This paper has demonstrated a hybrid feature-based approach to fish species classification based on segmented images of the Fish4Knowledge dataset. The results obtained have shown that the phenotypic features and the deep CNN embeddings are complementary to each other, and their combination yields a significantly higher accuracy than could be potentially achieved through the use of either feature set alone. Finally, the best-performing model, namely the hybrid feature set with Xception embeddings and an SVM classifier, achieved an accuracy of 97.42% and surpassed a number of recent approaches on fish species classification tasks. These results provide evidence that the combination of biologically meaningful features with state-of-the-art deep learning features is a robust and viable solution to underwater fish identification.

The main novelty of this study is the synergy that has been created between the U-Net-driven segmentation and the multi-classifier evaluation pipeline. Differently from previous studies that commonly used raw pixel data, our approach ensures that the features used by the SVM, Random Forest, and XGBoost classifiers are exclusively indicative of the fish's biological phenotype. Future work will focus on the development of transformer models, real-time implementation on edge devices, and the extension of the system to video-based fish tracking..

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